ChemRMBench: Auditing Cold Continuation Under Matched Actuator Inputs

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Abstract

The same feedback program need not emit identical future actions after an approximate state restoration. We introduce a private fixed-input control for a fresh-process laboratory continuation, recording exact setter tensors and actual SDK simulator-input buffers before every integration step. A fresh cold process restores at source cursor 2,400 and consumes only the remaining 11,027 commands plus 30 hold steps. We retain the original strict state and task-clock criteria and distinguish this diagnostic from autonomous feedback-policy recovery. The result below is one independently read engineering pair; no ordinary cold-coverage, memory or full-benchmark admission is added.

1. Matching programs versus matching applied inputs

A checkpoint may reproduce the observable state within its declared tolerance while its future feedback actions differ. Let the simulator evolve as

$$\begin{aligned}(x_{t+1}, h_{t+1}) &= F(x_t, h_t, a_t),\\ a_t &= (u_t, g(x_t; c_t)).\end{aligned} \tag{1}$$

where x_t is registered physical state, h_t is unexported solver state, u_t is the archived position-target command and g is the pinned feedback program. Restoring x_t approximately and the controller cursor c_t exactly does not imply exact equality of $g(x_t; c_t)$ across two trajectories. V99 retains its original strict negative result: seven geometric/program endpoint comparisons pass, but seven strict comparisons fail on an effort feedforward target field. This observation does not isolate h_t.

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Research progress. One private fixed-input pair; not submitted or peer reviewed.

V100 tests a separate private counterfactual,

$$a_t^{\mathrm{cold}} = a_t^{\mathrm{live}}, \qquad t = 2400, \ldots, 13456, \tag{2}$$

where the canonical step index includes thirty hold steps. A fresh live run records all ordered setter inputs as exact float32 tensors, effective target readbacks and SDK simulator-input buffers. A fresh cold process validates that sealed trace before restoration and consumes its 30 hold and 11,027 remaining command steps. It never re-executes the 2,400-command prefix. The hold's empty setter sequence is preserved instead of inserting redundant setter calls. Three target channels cover all joints of nine articulations; all other named asset setters and state/property writers are refused. Explicit actuator models and external wrenches are outside this instrument's scope. For the pinned implicit models, the audit verifies all three simulator input buffers before integration; it does not assert equal physical reaction forces produced by implicit PD or recover hidden solver caches.

For each endpoint s, the original strict criterion remains

$$G_s = G_{\mathrm{registry}} \wedge G_{\mathrm{host}} \wedge G_{\mathrm{geometry}}(\epsilon) \wedge G_{\mathrm{targets}}, \tag{3}$$

with the original 1 mm, 0.5° and 10^{-4} saved-joint-coordinate limits. Initial observed-velocity checks and thirty matched hold samples remain separate gates. The V99 additional 1 mm task-return condition is retained; it is distinguished from the historical 20 mm task-return limit.

The declared paused virtual task clock preserves the source prefix:

$$T^{\mathrm{cold}}(k) = T_{2400}^{\mathrm{saved}} + k\Delta t. \tag{4}$$

Hold steps are charged. Fresh-scene warmup and process downtime are excluded explicitly. Source-command alignment and every native/task clock row are recomputed independently. Movie presentation time is not the acceptance clock.

The complete action trace is private validator data, never an evaluated agent's future-action oracle. Even a passing V100 result is a fixed-input engineering control; ordinary cold restart coverage remains $1/8$. No causal memory gain, continuous chemical timing, learned policy, independent random task repetition or full-benchmark acceptance follows from this pair.

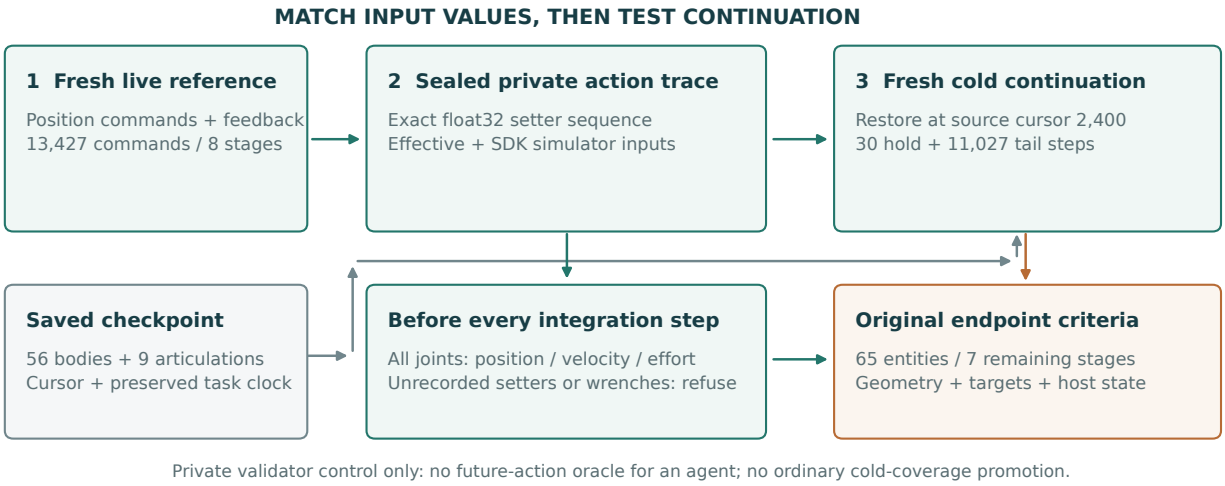


Figure 1. V100 explanatory control architecture, not a sensor image. The private action trace binds the live/cold applied-input values. The checkpoint separately supplies registered physical state, source cursor and declared task clock. All original state criteria remain; private future actions are never exposed to an agent.

2. Measured fixed-input result

The original sealed reader reports `PASS` for this private fixed-input control. The fresh live reference completes 13,427 commands; the cold process completes 11,027 tail commands and 30 hold steps. All 11,057 cold integration inputs match the corresponding live inputs exactly, including simulator-target byte records for all nine articulations. Initial restoration, observed velocities and thirty matched hold samples pass. The original strict endpoint criterion passes 7/7 stages; the cold final return error is 0.353397 mm. Table 1 preserves each stage's result. The reader independently binds targets to the committed physical stage archives and checks every source/native/task-clock row. The two workers allocate at most 0.279668 GPU h; there are no model calls or training jobs. A local resource refusal and an omitted-launch-field failure precede this fresh R3 cohort; neither started an SDK worker or consumed allocated GPU hours. Both are retained rather than pooled as physical trials. This is one fixed-condition live/cold pair. Seven stage endpoints are not seven independent task samples. Ordinary feedback-policy cold coverage remains $1/8$, and V99's strict negative result is unchanged. A passing fixed-input control shows that strict agreement is achievable in this restricted counterfactual; it does not by itself attribute V99's divergence to one physical cause or validate autonomous feedback continuation.

Table 1. Original V100 endpoint criteria. Maxima cover all 65 registered entities. Saved joint coordinates use the original per-joint coordinates; the 10^{-4} bound is not uniformly an angular unit.

Stage	Position (mm)	Angle (°)	Strict
Grasp	0.185492	0.152504	Pass
Lift	0.238345	0.152504	Pass
Transport	0.259081	0.152504	Pass
Place/release	0.278311	0.152504	Pass
Regrasp	0.286381	0.152504	Pass
Return	0.349665	0.152504	Pass
Retreat	0.430962	0.152504	Pass

3. Next evidence boundary

A future feedback-policy test must keep initial checkpoint integrity separate from equality of subsequent feedback values. It must specify the controller, observation and clock contract prospectively and report its own evidence tier. V100 does not supply a new autonomous-policy pass or justify changing prior thresholds after seeing these outcomes. Calibrated chemical events, independent memory controls, task-space generalization and policy training remain open.